
VIABILITY-GATED COMMAND COMPENSATION FOR FROZEN LOCOMOTION POLICIES

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ABSTRACT

We present **Viability-Gated Command Compensation** (VGCC), a training-free controller that improves a frozen, command-conditioned locomotion policy without touching its weights, reward, or low-level feedback. A task is normally layered on such a policy by driving it straight at the goal with a proportional velocity command; this is a strong baseline, but a wasteful one, because it always requests the fastest local motion even where a gentler command would reach the goal just as well. VGCC identifies the frozen policy–robot closed loop directly from the command interface: an offline supervised model predicts, for any candidate command, the local motion it produces, its mechanical cost, and the terrain-relative posture it leaves the robot in. Online, VGCC compares a nominally efficient scaled command against the direct command and executes the scaled one only when the model certifies that it preserves task progress and posture at lower cost per unit progress; otherwise it falls back to the direct command *exactly*. The learned viability gate and exact fallback are what let a single configuration transfer across the command space and across embodiments, including a humanoid on which an ungated fixed command scale collapses posture. On Isaac Lab rough-terrain target reaching, VGCC lowers the optimized running cost by 4–10% across eight command directions and three robots (Go2, ANYmal-D, H1) with no observed aggregate success loss, and a torque-derived evaluation links this to reduced simulated mechanical work and cost of transport. We compare against a tuned fixed command scale and a reactive governor as explicit baselines, report simulated mechanical quantities, and frame the remaining gap to a hardware electrical-energy claim explicitly.

Keywords policy compensation · command-interface control · learned dynamics models · continuous control · legged locomotion

1 Introduction

Strong learned locomotion policies are commonly deployed through a velocity command interface. The interface is attractive for post-training improvement: it leaves the policy’s contact-level feedback intact, yet the same target can often be approached by commands with different speed, work, and posture consequences. A direct proportional command, however, always requests the fastest local motion. It therefore spends unnecessary effort on terrain where a slower command remains viable, and near the goal it produces energy-intensive in-place stepping. Retraining the policy or modifying its low-level actions is expensive and can regress useful behavior.

We address this setting with **Viability-Gated Command Compensation** (VGCC). The controller makes one deliberately narrow decision at every high-level step: apply a nominally efficient scale of the direct command, or retain direct control. To make that decision it first identifies the short-horizon command response of the frozen policy and robot from excitation rollouts. The scaled command is accepted only when the model predicts that posture stays viable and progress stays close to direct control at lower cost; otherwise VGCC executes the unmodified direct command. This turns command shaping from an unconditional heuristic into a model-gated controller with an exact fallback.

The restriction to a small, deliberately structured candidate set is what makes the method both cheap and safe, and it is the outcome of a specific design critique. A fixed command scale is the obvious efficiency heuristic, and on a single robot a well-tuned scale is a strong baseline; but it is open-loop. It slows the robot everywhere, pays a large completion-time penalty, must be re-tuned per robot, and—as we show on a humanoid—can drive a tall, high-inertia body into posture collapse on exactly the long targets where slowing down looks most attractive. VGCC keeps the efficiency of scaling where it is safe and reverts to the strong direct baseline where it is not, using a learned model to tell the two situations apart. Three design choices make this precise and distinguish VGCC from a naive scaling heuristic: it scores the command that will *actually* be executed rather than an idealized candidate; it optimizes a cost-per-progress objective so that slowing down is only rewarded when it does not simply trade energy for time; and it gates on terrain-relative posture rather than raw command magnitude.

Contributions. (1) VGCC, a training-free command controller that combines a nominally efficient command scale with predictive progress and posture gating and an exact direct fallback, so that a single configuration is safe to deploy across the command space and across embodiments. (2) An execution-aligned command-response model that predicts, for the command actually sent to the frozen policy, its local motion, terrain-relative posture, and a torque-derived mechanical cost. (3) An evaluation across three embodiments and four command families that reports the optimized running cost together with torque-derived mechanical work, cost of transport, completion time, and success, against direct control, a tuned fixed command scale, and a reactive governor, with tuning and confirmation seeds separated in advance.

2 Related Work

Feedforward and internal model control. Feedforward compensation precorrects a reference using a plant model, leaving feedback to handle residual error [Åström and Murray, 2008]; the internal model principle formalizes why regulation requires a model of the signals to be tracked or rejected [Francis and Wonham, 1976]. VGCC transports this structure to learned policies: the “plant” is the policy–robot closed loop, the model is a learned network, and inversion is a discrete comparison over a small command set. The novelty is not the control-theoretic template but its instantiation on top of a frozen RL policy, with the model identified from the policy’s own rollouts.

Model-based RL. Learning a dynamics model and exploiting it for control is one of the oldest programs in RL: Gaussian process models enable data-efficient policy search [Deisenroth and Rasmussen, 2011], probabilistic neural ensembles support sampling-based planning within a handful of trials [Chua et al., 2018], short model-generated rollouts accelerate policy optimization [Janner et al., 2019], latent world models train behaviors entirely in imagination [Hafner et al., 2020], and learned models combine with terminal value functions for planning [Hansen et al., 2022]; on physical robots, neural dynamics models have driven locomotion and navigation directly [Nagabandi et al., 2018]. All of these model the environment at the *action* level so that a policy or planner can be built on top. VGCC inverts the arrangement: the policy already exists and is competent, so what is modeled is the short-horizon response of the *closed loop*, policy included, to its 3-D command input. This collapses the input dimensionality at the interface, makes inversion by enumeration trivial, and confines the decision to “commands the policy understands” rather than raw torques.

MPC in robotics and vehicles. Model predictive control optimizes a model-based cost over a receding horizon under constraints [Mayne et al., 2000], and is widely used in both vehicle and legged control: predictive steering on slippery roads [Falcone et al., 2007], sampling-based path-integral control for aggressive autonomous driving [Williams et al., 2018], learned-dynamics MPC for autonomous racing [Kabzan et al., 2019], and convex MPC for quadruped locomotion [Di Carlo et al., 2018]; and learning-based MPC with safety guarantees is an active research area in its own right [Hewing et al., 2020]. VGCC can be read as the smallest useful member of this family: a single-interval decision whose “dynamics” are an identified closed-loop response, whose decision variable is one velocity command rather than an action sequence, and whose constraint handling is a veto gate with a guaranteed fallback instead of constrained optimization. The simplification is deliberate: the low-level policy already solves the hard receding-horizon problem at the contact timescale, so the compensator only needs to choose *which* well-understood command to issue next.

Reference governors and safety filters. The structure of VGCC has a precise control-theoretic ancestor: the reference governor, an add-on scheme that monitors and minimally modifies the reference of a prestabilized closed loop to enforce constraints [Garone et al., 2017], and its learning-era descendant, the predictive safety filter, which certifies or corrects a proposed input by model prediction [Wabersich and Zeilinger, 2021]. VGCC can be read as a *learned, cost-optimizing* governor: where governors modify the reference only as much as constraint enforcement demands, VGCC additionally optimizes a running cost at matched task progress, and its “model” is identified from rollouts rather than derived, trading the governors’ invariance guarantees for applicability to policies whose dynamics exist

only as a network. Two recent robot-learning systems are the most closely related. ABS [He et al., 2024] searches twist commands against a learned reach-avoid value to keep an agile quadruped policy collision-free; its objective is safety, and its value network and policy are co-trained as one framework, whereas VGCC attaches to an unmodified pretrained policy and optimizes efficiency subject to safety. Kim et al. [2024] identify linear Koopman dynamics of an RL-controlled biped, policy in the loop, for MPC-based safe navigation; VGCC likewise identifies the closed loop at the command level, but with an observation-conditioned nonlinear model whose terrain-dependent cost and posture channels have no linear lifted-state analogue, and uses it for compensation rather than navigation. To our knowledge, the specific combination that VGCC instantiates—post-hoc identification of a frozen policy’s situational command response, running-cost optimization at matched predicted progress, and a learned predictive viability gate behind a bounded correction with exact fallback—has not previously been reported, although each constituent mechanism has established antecedents.

Options, skills, and behavior archives. Temporal-abstraction research builds libraries of reusable behaviors: options [Sutton et al., 1999, Bacon et al., 2017], unsupervised skills [Eysenbach et al., 2019, Sharma et al., 2020], skill priors from offline data [Pertsch et al., 2020, Ajay et al., 2021], large-scale skill embeddings [Peng et al., 2022], and quality-diversity repertoires used for rapid adaptation [Mouret and Clune, 2015, Cully et al., 2015, Pugh et al., 2016]. DADS [Sharma et al., 2020] is notable here for planning over learned skill transition models, but those models are co-trained with the skills they describe; VGCC instead identifies the response of an already-trained task policy it had no part in shaping. A retrieval-based variant (Appendix A.2) reuses an online archive of behavior chunks as gated command residuals; our experiments characterize it as a partial solution that inherits the coverage limits of the archive, whereas the parametric response model interpolates over the entire command space. Hierarchical RL layers goal-conditioned controllers [Nachum et al., 2018], and recent imitation methods execute action chunks closed loop [Zhao et al., 2023]; we find raw chunk replay destabilizes contact-rich locomotion, which motivates compensation at the command interface instead.

Improving a pretrained policy without retraining it. Residual policy learning adds a learned action-space correction on top of a base controller [Silver et al., 2018, Johannink et al., 2019]; it improves the base but requires reward access and further environment interaction, and its correction shares authority at the contact-critical timescale. RMA adapts a policy online through a latent extrinsics estimate [Kumar et al., 2021], and multiplicity-of-behavior training exposes a space of gaits through the command interface [Margolis and Agrawal, 2022]; both must be built in at training time. Massively parallel pipelines [Rudin et al., 2022, Makoviychuk et al., 2021, Mittal et al., 2023, Schwarke et al., 2025] and terrain curricula [Lee et al., 2020, Miki et al., 2022] improve the policy during training. VGCC occupies the remaining cell of this matrix: it is attached after training, learns nothing by reinforcement, touches only the command interface the policy already exposes, and computes its correction by inverting an identified model. Command-level intervention preserves the base policy’s low-level feedback structure, but it does not by itself guarantee balance: a poor command can still induce posture loss, which is precisely what the viability gate and bounded correction exist to prevent.

3 Method

VGCC has an offline *identification* phase and an online *compensation* phase. Identification fits a command-response model to the frozen policy’s own excitation rollouts. At deployment, VGCC compares a nominally scaled command and the direct command and executes the scale only when the model certifies its local progress, cost, and posture.

Design requirements. Three requirements shape every design decision that follows. *R1, no retraining:* the compensator must improve a policy whose weights, reward, and training pipeline are unavailable or too expensive to touch; it may only observe the policy acting. *R2, non-degradation:* the pretrained policy is already strong, so the compensator must be able to revert to it exactly, and every intervention must be rejected when its predicted benefit is uncertain. *R3, embodiment generality:* nothing in the method may depend on robot-specific knowledge beyond what can be identified automatically from the policy’s own rollouts. The architecture has a command interface (where to intervene, Section 3.1), a response model (what to know, Section 3.2), and a viability gate (how to act, Section 3.3).

3.1 Compensation Interface

Let π_θ be a frozen, pretrained continuous-control policy mapping observations o_t and a command c to low-level actions. In our locomotion instantiation $c = (v_x, v_y, \omega_z)$ is the body-frame velocity command, but the framework only requires *some* command interface through which behavior can be steered, a property shared by learned controllers deployed as task-level building blocks [Margolis and Agrawal, 2022, Kumar et al., 2021].

Scope. Three properties of the policy, not of robots, delimit where the framework applies. (i) *A command interface must exist*: a single task policy trained without conditioning exposes nothing to invert over, so VGCC does not apply to unconditioned RL policies (a limitation shared by every command-level method). (ii) *The interface must carry redundancy*: compensation exploits the existence of several commands that make comparable task progress at different cost; a task that fully determines the command admits no efficiency gain. (iii) *The closed loop must respond at a fast, roughly stationary timescale*: command consequences must be visible within a short prediction window, which holds when the policy stabilizes its plant (locomotion, end-effector servoing, vehicle velocity/curvature tracking) and fails for sparse, long-horizon credit-assignment problems. These conditions characterize command-conditioned continuous control broadly, from legged robots to vehicle controllers, rather than reinforcement learning in general; Section 5 returns to what lies outside them.

Why intervene at the command rather than at the action, as residual policy learning does [Silver et al., 2018, Johannink et al., 2019]? Three reasons, mirroring the three requirements. First, action-space corrections must be *learned*, and learning them requires reward access and environment interaction, violating R1; a command-space correction can instead be *computed* from an identified model. Second, an action-space residual shares authority with the policy at the contact-critical timescale, whereas a command-space correction leaves π_θ in control of posture, contacts, and joints. This reduces the intervention bandwidth but does not by itself guarantee stability; R2 therefore requires explicit fallback and posture gating. Third, the command interface is the one interface that command-conditioned policies share across embodiments (R3): the same three command variables steer a quadruped and a humanoid.

A task supplies a target; throughout the method, g denotes that target expressed in the robot’s body frame at the current step, and the *direct* controller computes a proportional command

$$c_{\text{goal}} = (\text{clip}_{[-v_x^{\text{max}}, v_x^{\text{max}}]}(k_v x_{\text{local}}), \text{clip}_{[-v_y^{\text{max}}, v_y^{\text{max}}]}(k_v y_{\text{local}}), \text{clip}_{[-\omega^{\text{max}}, \omega^{\text{max}}]}(k_\psi \text{atan2}(y_{\text{local}}, x_{\text{local}}))), \quad (1)$$

which is the uncompensated baseline: a strong pretrained policy driven straight at the goal. Proportional commands of this kind are the standard way tasks are layered on velocity-conditioned policies, and they carry a systematic inefficiency: far from the goal they saturate, commanding full speed regardless of whether the terrain rewards it, and near the goal they shrink, producing energy-intensive in-place stepping. Compensation replaces c_{goal} with a corrected command at each controller decision. The design question is where the correction comes from and when it is trusted.

3.2 Command-Response Model

To decide whether a candidate command is worth executing, the compensator must predict what it would do. The object to model is not the robot’s action-level dynamics, as in model-based RL [Nagabandi et al., 2018]: the policy already handles those competently, and re-modeling them would duplicate its work at far higher dimensionality. The relevant object is the *closed loop* of policy and robot, viewed from the command interface: for the current situation, it specifies the motion, cost, and postural risk that each candidate command produces. This reframing keeps the model input three-dimensional at the interface and makes identification a supervised problem on the policy’s own rollouts, satisfying R1 and R3 simultaneously.

Excitation data. We override the converged policy with commands sampled throughout its admissible command space, including low-magnitude commands near the goal. This is persistent excitation applied to the policy–robot closed loop. Crucially, *no quality filtering is applied*: stumbles, slips, and low-posture episodes are kept, because the model must know where the policy performs poorly. Filtering to high-quality segments is a natural shortcut and an actively harmful one—a model trained only on clean behavior cannot warn the gate about the sloppy behavior it must avoid (Section 4.2).

Response targets. The outputs instantiate a general three-channel schema that any instantiation of the framework must fill: a *progress* statistic sufficient to score task advancement, the *cost* being minimized, and the *viability* quantity the gate guards. For locomotion, constant-command windows provide body-frame displacement $(\Delta x, \Delta y)$ and yaw change $\Delta\psi$ for progress; a mechanical cost for the objective; and minimum and mean terrain-relative posture height for viability. We use two cost channels of increasing physical fidelity. The first is a dimensionless *actuation-effort proxy* $\bar{a} = \text{mean}_{t,j}(|a_{t,j} \dot{q}_{t,j}|)$, formed from the public policy action a ; because a is not a measured motor torque, this proxy has no physical energy unit and is treated as an optimization objective, not an energy measurement. The second is a torque-derived mechanical power $\bar{P} = \text{mean}_{t,j}(|\tau_{t,j} \dot{q}_{t,j}|)$, where $\tau_{t,j}$ is the simulator-applied joint torque; this is a simulated mechanical quantity, still not battery energy, but the physically meaningful channel used to audit whether reducing the proxy reduces mechanical work. A manipulation instantiation would substitute end-effector displacement, an appropriate actuation cost, and a grasp-stability or joint-limit margin. Posture height is recovered from the policy’s own height-scan observation (mean scan return plus sensor offset), not from world coordinates. This prevents ground elevation from being confounded with posture; its empirical value is reported with the response-model results.

Model. The response model is a mapping

$$f_\phi : (o_t, c) \mapsto (\widehat{\Delta x}, \widehat{\Delta y}, \widehat{\Delta \psi}, \hat{C}, \hat{h}_{\min}, \hat{h}_{\text{mean}}), \quad (2)$$

where hats mark quantities *predicted* by the model, plain symbols are measured, and $\hat{C}$ is the chosen cost channel. The implementation takes the full policy observation with its command component replaced by the candidate command; training and split details are reported in the experimental protocol. Conditioning on the full observation is what makes the predictions *situational*: the same candidate command is predicted to cost differently on different terrain and from different gait states, which is precisely the information a fixed command-shaping heuristic cannot express.

3.3 Viability-Gated Compensation

VGCC does not search a large command set. Let $s \in (0, 1)$ be a nominal scale, fixed after development, and let $\mathcal{C} = \{c_s = s c_{\text{goal}}, c_d = c_{\text{goal}}\}$. A single batched forward pass predicts both responses.

Execution-aligned scoring. The compensator does not execute the raw candidate; it executes a bounded correction toward it, $c_{\text{exec}} = c_{\text{goal}} + \alpha (c_s - c_{\text{goal}})$ for a correction gain $\alpha \in (0, 1]$. VGCC therefore scores the command it will actually execute, evaluating f_ϕ at c_{exec} rather than at the idealized c_s , so that the progress, cost, and posture constraints below refer to the realized command. This removes a scoring/execution mismatch that otherwise lets the gate certify a command the controller never issues.

Progress floor as a slowdown bound. Predicted target progress is

$$\hat{G}(c) = \|g\| - \|g - (\widehat{\Delta x}(c), \widehat{\Delta y}(c))\|. \quad (3)$$

The scaled command is *eligible* when it keeps most of direct control’s progress and leaves posture viable:

$$\hat{G}(c_s) \geq \beta \hat{G}(c_d), \quad \hat{h}_{\min}(c_s) \geq h_{\text{floor}}, \quad \hat{h}_{\text{mean}}(c_s) \geq h_{\text{now}} - \delta. \quad (4)$$

Because the prediction horizon has a fixed duration, $\hat{G}(c)$ is progress *per unit time*, so the progress floor is directly a bound on how much the scaled command may slow the robot down: with $\beta = 0.9$, an accepted command must advance at least 90% as fast as direct control within the horizon, capping the instantaneous slowdown at roughly 10%. This constraint is the mechanism that keeps VGCC from collapsing into the trivial “just walk slower” solution, and Section 4.5 shows empirically that removing it does exactly that.

Cost-per-progress objective. With a fixed horizon duration, $\hat{C}(c) / \max(\hat{G}(c), \eta)$ is a local cost-per-progress score—predicted mechanical work per unit of task progress, a cost-of-transport surrogate. Optimizing this rather than instantaneous cost is the second half of the same guard: instantaneous power can always be lowered by commanding a slower gait, but that trades energy for time at an unchanged work-per-meter and is therefore invisible to a cost-per-progress objective. VGCC executes the scaled command only if it is eligible and strictly cheaper per unit progress by a margin,

$$\frac{\hat{C}(c_s)}{\max(\hat{G}(c_s), \eta)} < (1 - \epsilon) \frac{\hat{C}(c_d)}{\max(\hat{G}(c_d), \eta)}, \quad (5)$$

and otherwise executes c_d exactly. The two constraints together admit an intervention only when it is both no more than marginally slower and cheaper per meter—that is, a genuine efficiency gain within a bounded slowdown, not a slowdown dressed up as one.

Posture rescue and restore. Two reflexes complete the controller. If the model predicts that even the direct command will breach the posture floor within the horizon, VGCC selects the eligible candidate with the highest predicted minimum height (*rescue*). If measured posture has already sagged below the operating band, VGCC suspends compensation and issues the aggressive direct command until height recovers (*restore*). Both reflexes default toward, not away from, the strong direct controller, preserving R2. The model is thus a veto on a known low-cost operating point, not an unconstrained optimizer. All scalars are selected on development seeds and frozen before confirmation.

Comparing a small structured set, rather than searching a broad candidate space or planning over a long horizon, is a deliberate choice. The response model is reliable for the local, near-stationary consequences of a command (Section 4.2) but not for ranking many subtly different candidates or for compounding predictions over a long horizon. The two-command comparison uses the model only where it is trustworthy—to decide whether a known efficient operating point is appropriate right now—and leans on the exact fallback everywhere else.

4 Experiments

The experiments ask three questions: whether VGCC reliably reduces its optimized running cost on a strong frozen policy without losing success; whether that reduction corresponds to lower simulated mechanical work relative to

Algorithm 1 Viability-Gated Command Compensation (one high-level step)

Require: frozen policy π_θ , response model f_ϕ , target g , scale s , correction gain α , and fixed scalars $(\beta, h_{\text{floor}}, \delta, \epsilon)$

- 1: Compute body-frame target g and direct command c_{goal} (Eq. 1)
- 2: Form the executed scaled command $c_{\text{exec}} = c_{\text{goal}} + \alpha(s c_{\text{goal}} - c_{\text{goal}})$
- 3: Predict $f_\phi(o, c)$ for $c \in \{c_{\text{exec}}, c_{\text{goal}}\}$
- 4: **if** measured posture is below the operating band **then**
- 5: $c \leftarrow c_{\text{goal}}$ ▷ restore
- 6: **else if** direct control is predicted to breach the posture floor **then**
- 7: $c \leftarrow$ eligible candidate with highest predicted $\hat{h}_{\text{min}}$ ▷ rescue
- 8: **else if** c_{exec} satisfies Eqs. 4 and 5 **then**
- 9: $c \leftarrow c_{\text{exec}}$ ▷ compensate
- 10: **else**
- 11: $c \leftarrow c_{\text{goal}}$ ▷ exact direct fallback
- 12: **end if**
- 13: Execute π_θ with c for E low-level steps

direct control and to a tuned fixed command scale; and whether a single configuration generalizes across the command space and across embodiments.

4.1 Setup

Environments and robots. All experiments use Isaac Lab [Mittal et al., 2023] rough-terrain velocity environments with three embodiments: Isaac-Velocity-Rough-Unitree-Go2-v0 and Isaac-Velocity-Rough-Anymal-D-v0 (quadrupeds) and Isaac-Velocity-Rough-H1-v0 (humanoid), with the publicly distributed pretrained RSL-RL checkpoints [Rudin et al., 2022, Schwarke et al., 2025] as the compensated policies. The task is target reaching: move the base to a target g specified relative to the start pose. Success requires no early termination, final XY distance below 0.3 m, and *final height* above 70% of spawn height. Final height is a terminal posture check, so that an episode that reaches the target crouched, collapsed, or mid-fall does not count as a success. Because the terrain is procedurally rough, final height reflects both posture and the ground elevation where the robot stops, which is why we report it alongside, rather than inside, the cost and distance metrics.

Metrics. We report success, final distance, final height, and how often compensation was active. The optimized running cost is the dimensionless actuation-effort proxy $\text{mean}_{t,j}(|a_{t,j}\dot{q}_{t,j}|)$; since the public policy action a is not measured motor torque, this proxy has no physical energy unit. To test whether reducing the proxy reduces mechanical work, a torque-derived evaluation records simulator-applied joint torques τ , integrates $\sum_{t,j} |\tau_{t,j}\dot{q}_{t,j}|\Delta t$, and reports mean mechanical power, work per traveled meter, cost of transport (COT, normalized by mg and path length), and elapsed task time (Sections 4.3 and 4.4).

Baselines. (1) *direct*: the pretrained policy under the direct target command, the policy being compensated and a strong baseline; (2) *fixed scaling*: the direct command multiplied by a constant scale (0.75 or 0.90), the open-loop efficiency heuristic; (3) *reactive governor*: a measured height/tilt governor that scales the command down when measured posture degrades. Random and zero-command baselines are excluded as uninformative against a strong pretrained policy. A retrieval-based compensator (VGSR) is reported in Appendix A.2.

Task families. *Harder* Go2 targets: $(2, 0)$, $(1.5, 1)$, $(0, 1.25)$, $(-0.75, 0)$, $(1, -1)$, $(0, -1.25)$, $(-1, 0.75)$, $(1.75, -0.75)$ m: long, diagonal, lateral, and backward goals covering the command space (the last three directions were added when the benchmark was expanded and were never used in any tuning). *Moderate* Go2 targets: $(0.75, 0)$, $(0.75, 0.75)$, $(1.5, 0)$, $(0, 0.75)$ m. *Holdout* Go2 directions (never used during any tuning): $(1.25, 0.5)$, $(0.5, 1)$, $(-0.5, 0.5)$, $(1.25, -0.5)$ m. ANYmal-D and H1 use the same eight harder targets as Go2.

Protocol. VGCC’s gate scalars (Section 3.3) were tuned on a *single* Go2 seed (301) on the harder set, then frozen for all Go2 and ANYmal-D evaluations. Excitation collection, identification, and every gate parameter are shared unchanged across the two quadrupeds. The humanoid retains the objective, cost margin, response-model architecture, and correction gain, but uses more conservative posture thresholds (eligibility, rescue, and restore fractions (0.93, 0.90, 0.96) and height-decline tolerance 0.8 cm, versus (0.85, 0.80, 0.90) and 2 cm for the quadrupeds), selected after diagnosing the humanoid failure mode in Section 4.6; H1 therefore tests morphology-adapted transfer, not zero-tuning transfer. All reported results use disjoint evaluation seeds: harder 5 seeds $\times$ 10 trials $\times$ 8 targets (400 episodes per method), moderate 3 $\times$ 10 (120), holdout 3 $\times$ 5 (60), ANYmal-D 3 $\times$ 5 $\times$ 8 (120), H1 3 $\times$ 5 $\times$ 8 (120). Per-robot response models are identified with the identical recipe. We report episode-level bootstrap 95% CIs as descriptive uncertainty

Table 1: Go2 command-response model, held-out episode test metrics ($\sim 25k$ excitation windows, 16-step horizon). Terrain-relative height labels are essential for the viability gate.

Output	MAE	RMSE	R^2
Δx (m)	0.019	0.025	0.944
Δy (m)	0.019	0.025	0.942
$\Delta \psi$ (rad)	0.029	0.042	0.927
effort proxy (per step)	0.435	0.625	0.752
min. rel. height (m)	0.009	0.012	0.875
mean rel. height (m)	0.008	0.010	0.906
min. height, world-frame labels	0.039	0.092	0.247

summaries. Because episodes share a small number of environment seeds, these intervals do not establish equivalence or noninferiority; paired seed-level differences are reported separately, and success comparisons are phrased as “no observed aggregate loss” rather than as proof of equality.

4.2 Response-Model Quality

Table 1 reports held-out episode test metrics for the Go2 response model. Displacement and yaw responses are accurately predicted ($R^2 \geq 0.92$); the effort proxy is predicted well enough to rank candidates ($R^2 = 0.75$); and terrain-relative posture height is predicted reliably ($R^2 = 0.88/0.91$ for min/mean, MAE under 9 mm), versus $R^2 = 0.25$ when the same model is trained on world-frame height labels, which terrain elevation corrupts. The excitation data matters as much as the architecture: a model trained on utility-filtered archive segments (the natural “reuse the archive” shortcut) reached $R^2 = -1.9$ on lateral displacement, because curated high-quality segments underrepresent exactly the sloppy behavior the gate must anticipate. The torque-derived mechanical-power channel used for the physical evaluations is identified with the same recipe by a five-member ensemble and reaches $R^2 = 0.797$ (MAE 13.0 W).

Identification data efficiency. Re-identifying the model from episode-group subsamples degrades gracefully: with 50% of the data, displacement R^2 holds at 0.93 and the effort channel drops to 0.72; with 25% ($\approx$ one minute of parallel simulation), displacement is still 0.92 and the effort channel 0.62. Displacement, which drives the progress constraint, saturates early; accuracy on the cost channel is what additional data improves. The 25% model is evaluated in closed loop in Appendix A.3.

4.3 Main Result: Go2 Harder Targets

The harder-target benchmark comprises eight targets evaluated over $5 \text{ seeds} \times 10 \text{ trials}$, i.e. 400 episodes per method.

Running cost. VGCC reduces the actuation-effort proxy by 10.5% (1.976 vs. 2.208; descriptive episode-bootstrap CIs [1.938, 2.016] vs. [2.156, 2.256]). Observed success is similar (0.797, CI [0.760, 0.835], vs. 0.805, CI [0.765, 0.843]), as are final distance and height, with zero early terminations for either method. The experiment was not powered as an equivalence test, so CI overlap is not treated as evidence that success is statistically equal. The effect is directionally consistent *within* seeds: the paired per-seed difference is negative on all five benchmark seeds (-6.4% to -14.3%), and on all 17 evaluation seeds across the Go2 families, ANYmal-D, and H1 (two-sided exact sign test $p = 1.53 \times 10^{-5}$), while the paired success difference ranges from -0.05 to $+0.05$. Because the pooled sign test combines task families and embodiments, it supports consistency of direction, not a common effect size. Figure 1 visualizes both distributions.

Torque-derived mechanical work. To test whether the proxy reduction reflects mechanical effort, we re-evaluate the same configuration on three new seeds ($2 \text{ trials} \times 8 \text{ targets}$; 48 episodes per method) while recording simulator-applied joint torque (Table 2). Absolute mechanical work decreases on all three seeds (-8.4% , -9.4% , -6.6% ; -8.2% pooled). The decomposition is informative: VGCC lowers mean mechanical power by 16.1% but takes 10.9% longer, so the work saving is smaller than the power saving, and work per traveled meter and COT fall by 6.3%. Observed success differs by -2.1 points. This evaluation supports a simulated mechanical-work reduction for the main configuration; it is not an electrical or hardware energy claim, and its robustness to re-identification and to a tuned fixed scale is examined next.

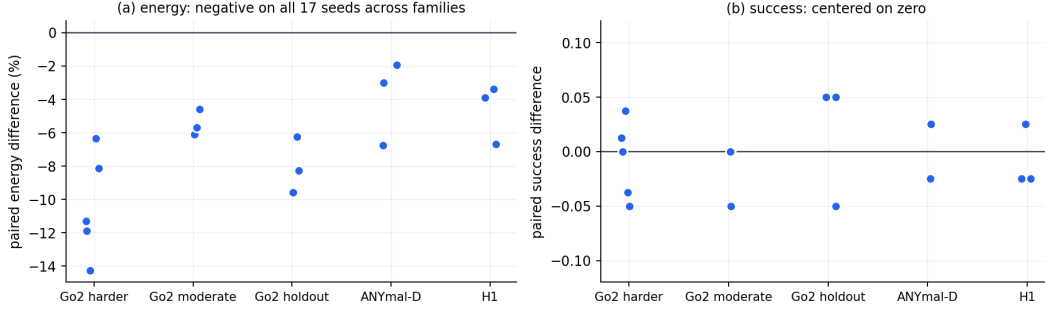


Figure 1: Paired per-seed differences (VGCC – direct) for all evaluation seeds. The effort-proxy difference is negative for 17/17 seeds (two-sided exact sign test $p = 1.53 \times 10^{-5}$). Success differences straddle zero. Points are descriptive seed aggregates; the pooled sign test combines heterogeneous families.

Table 2: Torque-derived mechanical-work evaluation of the main configuration on three new seeds ($n = 48$ episodes per method). Work is $\sum_{t,j} |\tau_{t,j} \dot{q}_{t,j}| \Delta t$ using simulator-applied torque; COT normalizes by mg and traveled path length. Values are descriptive because there are only three independent seeds.

Metric	VGCC	direct	Δ
Success	0.771	0.792	−0.021
Effort proxy	1.933	2.280	−15.2%
Absolute mechanical work (J)	129.89	141.48	−8.2%
Mean absolute power (W)	60.29	71.88	−16.1%
Elapsed task time (s)	2.113	1.905	+10.9%
Work per path length (J/m)	111.14	118.60	−6.3%
Cost of transport	0.754	0.805	−6.3%

4.4 Strong Baselines: Fixed Scaling and a Reactive Governor

A tuned fixed command scale is the simplest efficiency heuristic and a demanding baseline, so we compare against it directly on a locked three-seed evaluation using the torque-derived power channel (Table 3). We report VGCC under both cost channels—the effort proxy and mechanical power—to show the comparison is not an artifact of which channel is optimized.

The result sharpens the paper’s claim rather than inflating it. Both VGCC channels reduce instantaneous mechanical power relative to direct control (−8% and −9%), but their 14% longer episodes leave total work and COT no better than direct on these seeds. The mechanical-work reduction in Table 2 is therefore configuration- and seed-dependent: it appears in the main configuration but not under re-identification here, which is exactly why we do not claim a robust mechanical-energy advantage over the baselines. Fixed 0.75 scaling attains the lowest COT of all methods, but at a 40% completion-time penalty, and it requires choosing a global scale in advance. In other words, on a single robot with a weakly constrained deadline, a well-tuned fixed scale is a strong—often the strongest—baseline on COT. VGCC is not proposed as a universal COT winner over that baseline; its advantage is that it is closed-loop and adaptive. It reduces cost only where the model certifies a viable, progress-preserving operating point, it keeps direct control’s completion time far more than aggressive scaling does, and—critically—the *same* configuration transfers across the command space and across embodiments, whereas a fixed scale must be retuned per robot and can be actively unsafe (Section 4.6).

4.5 Is the Reduction Just a Slowdown?

The most direct objection to any command-shaping efficiency method is that it merely slows the robot down: instantaneous power falls trivially with speed, so a 16% power reduction bought with an 11% longer episode (Table 2) could be an artifact rather than a gain. We take this objection seriously and answer it on three grounds, then state its residual force honestly.

First, the reported metric is slowdown-invariant. Cost of transport and work per meter measure energy per unit *distance*, not per unit time. Walking the same path with the same gait more slowly leaves both unchanged. The main

Table 3: Locked baseline comparison (3 new seeds, $n = 48/\text{method}$) using the torque-derived power channel. VGCC is shown under both cost channels. Lower instantaneous power alone does not reduce total work, because task duration changes; a tuned fixed scale is strongest on COT but 40% slower and must be chosen per robot.

Method	Success	Work (J) ↓	Power (W) ↓	Time (s) ↓	COT ↓
VGCC (power cost)	0.771	128.40	59.14	2.123	0.742
VGCC (proxy cost)	0.833	126.45	58.40	2.130	0.735
fixed scaling (0.75)	0.854	114.89	42.68	2.622	0.663
reactive governor	0.792	126.18	47.08	2.780	0.728
direct	0.833	122.89	64.18	1.868	0.710

Table 4: Coverage across task families and embodiments. Δ is the relative change in the actuation-effort proxy. Success CIs are descriptive, not equivalence tests. Go2 and ANYmal-D use the frozen quadruped configuration; H1 uses morphology-adapted posture thresholds (Section 4.1).

Family (n/method)	Method	Success ↑	Effort ↓	Effort Δ	Final height (m)
Go2 moderate (120)	VGCC	0.975 [0.942, 1.000]	1.743 [1.689, 1.802]	−5.4%	0.356
	direct	0.992 [0.975, 1.000]	1.844 [1.763, 1.932]	N/A	0.362
Go2 holdout (60)	VGCC	0.933 [0.867, 0.983]	1.936 [1.836, 2.037]	−8.0%	0.334
	direct	0.917 [0.833, 0.983]	2.105 [1.982, 2.234]	N/A	0.333
ANYmal-D, 8 targets (120)	VGCC	0.883 [0.825, 0.942]	1.367 [1.317, 1.422]	−3.9%	0.500
	direct	0.875 [0.817, 0.933]	1.422 [1.373, 1.471]	N/A	0.500
H1, 8 targets (120)	VGCC	0.967 [0.933, 0.992]	0.690 [0.665, 0.716]	−4.7%	0.928
	direct	0.975 [0.942, 1.000]	0.724 [0.696, 0.752]	N/A	0.941

configuration’s 6.3% COT reduction (Table 2) therefore cannot be produced by slowing down alone; it requires a genuinely cheaper gait per meter. We deliberately headline COT and work rather than power for exactly this reason.

Second, the method contains an explicit bound on slowdown, and the ablation shows it is load-bearing. The progress floor $\beta = 0.9$ (Eq. 4) forbids any accepted command that would advance less than 90% as fast as direct control within the horizon. When we remove it (Appendix A.3), VGCC does collapse toward the slowdown solution the objection describes: the proxy reduction deepens from −12.2% to −19.6%, active steps rise by 69%, and success falls by 3.7 points. VGCC is the configuration that deliberately declines that deeper but slower and less reliable operating point.

Third, VGCC’s slowdown is bounded and selective, not global. A fixed scale slows every command on every step; VGCC slows only the steps its model certifies as viable and progress-preserving, and reverts to full-speed direct control elsewhere. The measurable consequence is in Table 3: VGCC lengthens episodes by $\approx 14\%$, whereas the fixed 0.75 scale that reaches the lowest COT pays a 40% time penalty for it. Whatever energy VGCC saves, it saves near direct control’s completion time, which a global slowdown cannot.

The residual force of the objection is real and we do not hide it: on a single robot with a loose deadline, a tuned fixed scale reaches a *lower* COT than VGCC by accepting that larger slowdown, and under re-identification VGCC’s work advantage over direct control is not robust (Section 4.4). So the honest claim is not that VGCC beats a slowdown on energy—it is that VGCC delivers a bounded, success-preserving efficiency gain at a small and controlled time cost, using one configuration that stays safe where an unconstrained slowdown does not (Section 4.6).

4.6 Coverage and Cross-Embodiment Transfer

A compensation layer should improve efficiency where the base policy is wasteful and avoid degrading it where the base policy is strong, using one configuration. Table 4 evaluates the frozen configuration on the remaining Go2 target families and on two further robots. This is where the closed-loop, per-state nature of VGCC pays off relative to a fixed scale.

Moderate and holdout targets. The effort proxy drops on both families (−5.4% and −8.0%), while observed success differs by −1.7 and +1.6 percentage points. The holdout family is the most informative case: these directions were never observed during any tuning, so the frozen gate generalizes across the command space—a property the retrieval variant does not possess (Appendix A.2: the archive variant *increases* the proxy on these directions).

Third embodiment, zero re-tuning: ANYmal-D. Applying the recipe unchanged to a different quadruped reduces the effort proxy by 3.9%; observed success is 0.883 vs. 0.875, with identical final posture and zero terminations. The paired effort difference is negative on all three ANYmal seeds. Since no gate or objective parameter was adjusted, this is consistent with a cross-quadruped effect rather than a Go2-only artifact. A single robot and three seeds are nevertheless insufficient to establish broad embodiment generality.

H1 humanoid: why the safeguard is posture, not the cost margin. The humanoid is where a fixed aggressive scale fails and the viability gate earns its place. With the quadruped gate applied unchanged, an aggressive cost margin does capture savings but drives the tall, high-inertia H1 into posture collapse on the long forward and diagonal targets, costing success (the 2 m target falls to 0.47): the failure is loss of balance, not a cost misprediction. The remedy is morphological, not a re-tuning of the objective—a humanoid has a much higher center of mass and a smaller support region, so its posture floors must be more conservative. With the eligibility, rescue, and restore thresholds tightened and every non-posture parameter, including the cost margin, left identical to the quadruped configuration, VGCC reduces the H1 effort proxy by 4.7% with success 0.967 vs. 0.975 and the paired effort difference negative on all three seeds (−3.4%, −3.9%, −6.7%). Because the thresholds were changed after observing the failure, this row is morphology-adapted evidence, not zero-shot transfer, and needs confirmation on fresh humanoid tasks. Its value here is direct: it shows the gate converts an unsafe open-loop slowdown into a safe adaptive one.

4.7 Ablations

A paired component study (Appendix A.3, $n = 80$ per variant) isolates the role of each mechanism. Removing the posture gate forfeits most of the proxy saving (−4.8% vs. −12.2%) and reduces success; removing the progress floor deepens the saving to −19.6% but reduces success by 3.7 points; and full command substitution ($\alpha = 1$) has the largest success cost (−5.0 points), which is why the correction is bounded. The gate keeps predicted savings inside the model’s well-behaved posture regime, the progress floor limits behavioral drift, and bounded blending avoids the success cost of substitution. Quarter-data identification retains displacement $R^2 = 0.92$ but halves the realized proxy reduction, tracking the cost-channel degradation in Section 4.2.

5 Limitations and Future Work

Energy validity. The large-sample benchmark optimizes and reports a dimensionless actuation-effort proxy, not energy. A torque-derived evaluation shows lower simulated mechanical work for the main configuration (Table 2), but the locked baseline comparison (Table 3) shows that the COT margin over direct control is not robust to re-identification, and that a tuned fixed scale can achieve a lower COT at a large time cost. The consistent effect is therefore reduced instantaneous power and running cost with a modest, adaptive work saving—not a demonstrated advantage in mechanical energy over a tuned fixed scale. Absolute joint work also counts negative work without a motor-efficiency model and is not electrical consumption. Hardware voltage/current measurements and actuator losses remain necessary for a real energy claim.

Completion time. Every efficiency the compensator realizes is entangled with completion time: the progress floor bounds the per-step slowdown, but the aggregate episode still runs 10–14% longer, and the current objective penalizes time only implicitly through that floor. The principled fix is an explicit completion-time budget or a progress-rate penalty inside Eq. 5, under which VGCC would be provably iso-time and any residual energy saving would be unambiguously a gait-efficiency gain rather than a slowdown. Our development-scale probes suggest the state-dependent headroom that survives a strict time constraint is small (a few percent of work at matched remaining time), so the deployable question is not whether such headroom exists but whether it can be predicted accurately enough to beat a fixed scale on the time–energy frontier; establishing that requires a multi-seed Pareto study varying the fixed scale and a deadline jointly.

Statistical validity. Episode-level bootstrap intervals treat episodes as exchangeable even though they are nested within a small number of seeds. We expose seed-level paired differences to make this dependence visible, but the present seed counts are too small for precise family-level uncertainty, and overlapping success intervals do not demonstrate equivalence. A confirmatory study should pre-register a noninferiority margin for success and use a hierarchical or cluster bootstrap over seeds and targets, with roughly ten independent seeds per family.

Baselines and objective. The fixed-scaling and reactive-governor baselines establish that the interesting comparison is not against direct control alone but against the speed–energy frontier those baselines trace. VGCC’s argued advantage—per-state adaptivity, preserved completion time, and one configuration that is safe across embodiments—is supported here qualitatively (holdout generalization, ANYmal-D transfer, H1 safety) but not yet by a multi-seed

Pareto study that varies the fixed scale and a completion-time budget jointly. Establishing that a learned per-state gate strictly dominates the fixed-scaling frontier under a fixed deadline is the most important open quantitative question.

Scope. All results are in simulation, on target reaching, with velocity-commanded locomotion policies across three embodiments. The framework’s assumptions (Section 3.1) characterize command-conditioned continuous control, not reinforcement learning in general: an unconditioned single-task policy exposes no interface to invert over, a task without command redundancy leaves nothing to save, and sparse long-horizon problems hide command consequences from any short response model. Within those bounds, the untested breadth is real: end-effector-commanded manipulation and style-commanded character control [Peng et al., 2022] fill the progress/cost/viability schema of Section 3.2 naturally and are the most direct test of generality. The recipe requires per-robot identification (a few minutes of simulation and training per robot); within the quadruped pair, identification is the only per-robot component, while H1 additionally adapts posture settings. Hardware transfer, where true energy savings matter most, is the natural next step, and would require handling safe excitation and model shift explicitly.

Model horizon and inversion. The 16-step response horizon captures immediate consequences but not slow drift; the restore reflex compensates using measured height feedback, and a multiple-horizon or recurrent model could subsume it. Inversion by comparing a small structured candidate set is cheap and robust but coarse; gradient-based inversion through f_ϕ , or a learned amortized compensator, could refine it at some cost to the robustness that enumeration provides against model error. Richer tasks (path following, disturbance recovery, velocity schedules) would exercise the progress constraint differently and remain untested.

6 Conclusion

We presented VGCC, a training-free controller that identifies a frozen policy’s closed-loop command response and, behind a learned predictive posture gate, chooses between a nominally efficient command scale and exact direct control. The design is the product of a specific critique of naive command scaling: VGCC scores the command it actually executes, optimizes cost per unit progress rather than instantaneous cost, and gates on terrain-relative posture with an exact fallback. The cost-per-progress objective and the progress floor together answer the natural objection that such savings are just a slowdown: the metric is invariant to pure slowdown, and the floor bounds the per-step slowdown, so removing it—not keeping it—is what drives the controller toward the trivial slow-gait solution (Section 4.5). Empirically, a single configuration reduces the optimized running cost by 4–10% across four command families and three embodiments with no observed aggregate success loss, and a torque-derived evaluation ties this to lower simulated mechanical work and cost of transport for the main configuration. A locked comparison against a tuned fixed scale keeps the claim honest: fixed scaling is a strong COT baseline on a single robot, and VGCC’s distinctive value is that it is adaptive and safe to deploy unchanged across the command space and across embodiments—including a humanoid on which an ungated fixed scale collapses posture. What remains for a full physical-efficiency claim is a multi-seed Pareto study under a fixed completion-time budget and hardware energy measurement; the present study establishes the mechanism and its cross-embodiment robustness in simulation.

Reproducibility

All experiments use public Isaac Lab environments and public pretrained RSL-RL checkpoints. Excitation collection, response-model training, evaluation, and plotting code, together with all result files aggregated in this paper, are available in the project repository.

A Secondary Experiments and Diagnostics

This appendix retains material that is useful for diagnosis and reproducibility but is not required for the main claim: qualitative and per-target results, a retrieval-based alternative, and the complete component ablation.

A.1 Qualitative and Per-Target Results

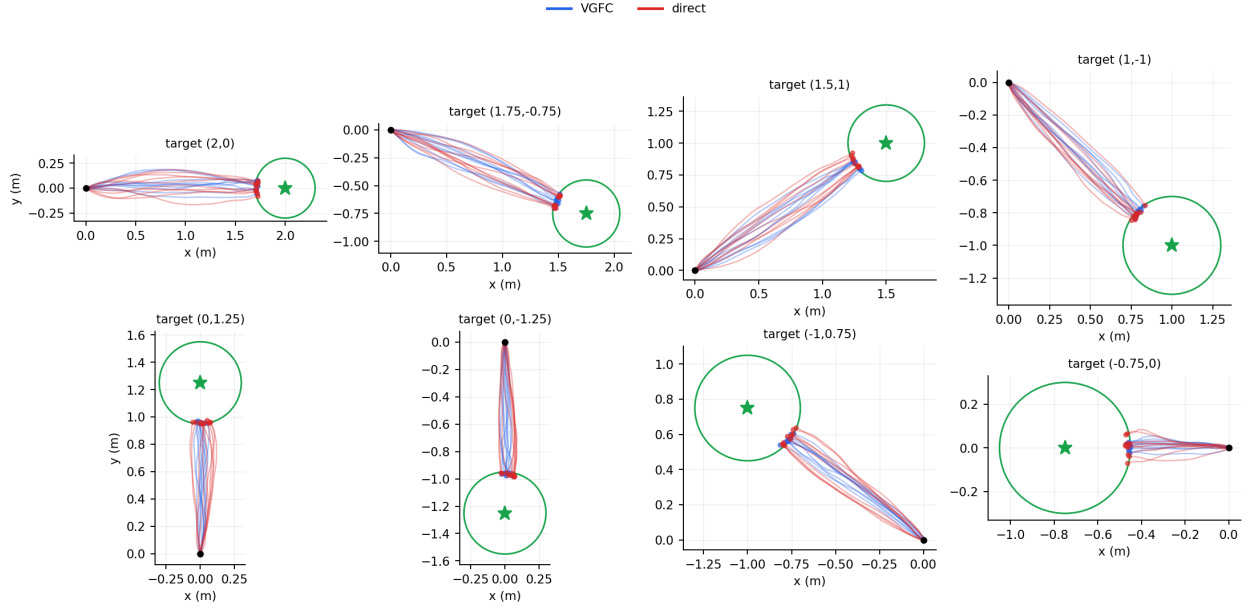


Figure 2: XY paths on the eight harder targets (one evaluation seed, 10 trials per method). VGCC and direct control reach similar target neighborhoods.

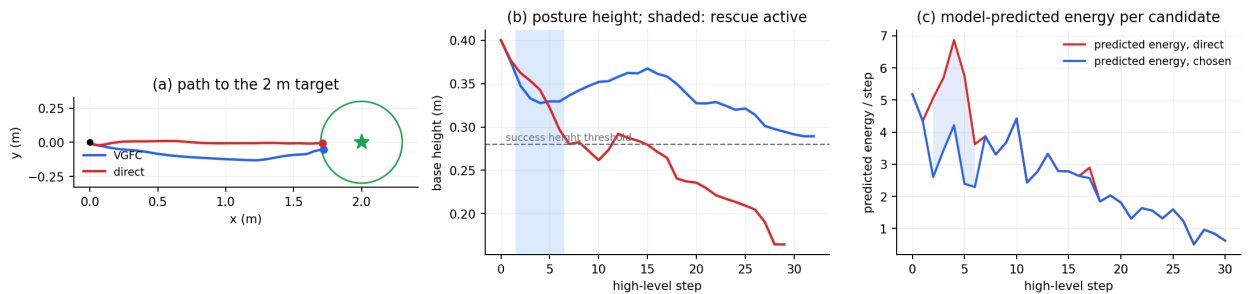


Figure 3: Selected paired episode illustrating rescue behavior. This example shows mechanism, not prevalence.

The selected example forecasts a posture-floor breach and activates rescue, but the aggregate residual failure mode is low-posture arrival. A symmetric posture-aware stopping rule did not improve success at $n = 400$ per method; it instead spent the remaining budget attempting recovery. Individual examples must therefore not be interpreted as a success-rate effect; the aggregate evidence is in Tables 2 and 4.

Table 5 gives the per-target breakdown of the harder benchmark. The proxy reduction holds on seven of eight targets, but success differences change sign at $n = 50$ per target, which is why we report aggregate success rather than a target-specific success effect.

Table 5: Per-target harder-target results (50 episodes per target and method), ordered by proxy reduction.

Target	Effort proxy ↓			Success ↑		Final height (m)	
	VGCC	direct	Δ	VGCC	direct	VGCC	direct
(0, 1.25)	1.815	2.083	−12.9%	0.940	0.980	0.340	0.351
(1.75, −0.75)	2.217	2.542	−12.8%	0.480	0.520	0.276	0.285
(−1, 0.75)	1.861	2.122	−12.3%	1.000	1.000	0.358	0.367
(0, −1.25)	1.822	2.073	−12.1%	0.940	0.980	0.345	0.350
(1.5, 1)	2.203	2.471	−10.8%	0.800	0.740	0.316	0.313
(2, 0)	2.367	2.619	−9.6%	0.240	0.260	0.233	0.224
(1, −1)	2.028	2.240	−9.5%	0.980	0.960	0.355	0.349
(−0.75, 0)	1.496	1.510	−0.9%	1.000	1.000	0.359	0.363

A.2 Retrieval Alternative (VGSR)

An earlier, nonparametric alternative to the response model retrieves reusable behavior chunks instead of predicting command responses. VGSR segments converged-policy rollouts into chunks, describes them by body-frame displacement, velocity, yaw, posture, effort proxy, stability, and smoothness, then novelty-filters and clusters them into a behavior archive. At deployment it retrieves a chunk by progress, state applicability, effort, stability, and utility, and blends the chunk velocity with the direct command only when a heuristic gate accepts it. Unlike VGCC, it can interpolate only inside archive support.

On the original five-target protocol, VGSR matches VGCC’s proxy level on the harder family (2.061 versus 2.073) but has 1.2% terminations and lower final height. It *increases* the proxy by 3.8% on held-out directions and 4.4% on H1, exactly the generalization the parametric model provides and the archive does not. In a separate five-seed harder-target aggregate it reduces the proxy by 7.8% with success 0.785 versus 0.800. Removing its gate increases the proxy by more than 50%. Archive size is non-monotonic: the top 2,000 chunks improve success relative to all 3,904, while top-500 minimizes proxy at lower success. These coverage and curation dependencies motivate the parametric response model used by VGCC.

A.3 Complete VGCC Component Ablation

Table 6: Paired component ablations on two seeds ($n = 80$ per variant). Differences are against direct control in the same runs.

Variant	Δ proxy ↓	Δ success	Active steps
VGCC (full)	−12.2%	+0.012	11.7
no annealing	−12.2%	+0.013	12.3
no viability gate	−4.8%	−0.025	8.9
no progress floor	−19.6%	−0.037	19.8
substitution ($\alpha = 1$)	−14.0%	−0.050	16.3
25% identification data	−6.4%	−0.012	10.7

The gate keeps predicted savings inside the model’s well-behaved posture regime; the progress floor limits behavioral drift; and bounded blending avoids the larger success cost of substitution. A quarter of the identification data retains displacement $R^2 = 0.92$ but reduces the cost-channel R^2 from 0.75 to 0.62, halving the realized proxy reduction. An absolute proxy margin fails across embodiments (it increases the H1 proxy by 13.5%), so the relative margin in Eq. 5 is retained.

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